7 Microcontrollers

Introduction

A microcontroller is a microcomputer. It along with I/O ports, timers/counters, memory, ADC/DAC, etc., is integrated on a single chip. It can be programmed to perform a specific operation or control a specific operation. A CPU built in a single VLSI chip is called a microprocessor. A digital computer having a microprocessor as the CPU, along with memory and I/O devices, is called a microcomputer.

Microprocessor versus Microcontroller

A computer-based system could be built using a single chip microcontroller with a minimum of external components. The figure given below shows the simplified block diagram of a microprocessor, which consists of the Arithmetic Logic Unit (ALU), general purpose registers, Stack Pointer (SP), Program Counter (PC), clock timing circuit, and interrupt circuit.

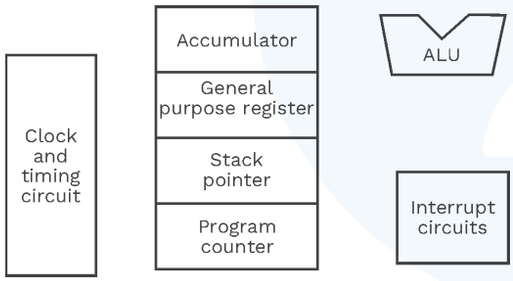


Fig. 7.1 Block Diagram of a Microprocessor

To make a computer system, not only should a microprocessor be sufficient but should also be required to add other peripherals, such as Read Only Memory (ROM), Read/Write Memory (RAM), decoders, drivers, and number of input/output devices required to make a complete microcomputer system. In addition, special purpose devices, such as interrupt controller, programmable timers, programmable I/O devices, and DMA controllers may be added to improve the capacity, performance, and flexibility of the microcomputer system.

The microcontroller incorporates all features that are found in a microprocessor. Besides this, it also has additional features to make a complete microcomputer system on its own. The microcontroller has built-in ROM, RAM, parallel I/O, serial I/O, counters, and clock circuit. The figure given below shows the simplified block diagram of a microcontroller.

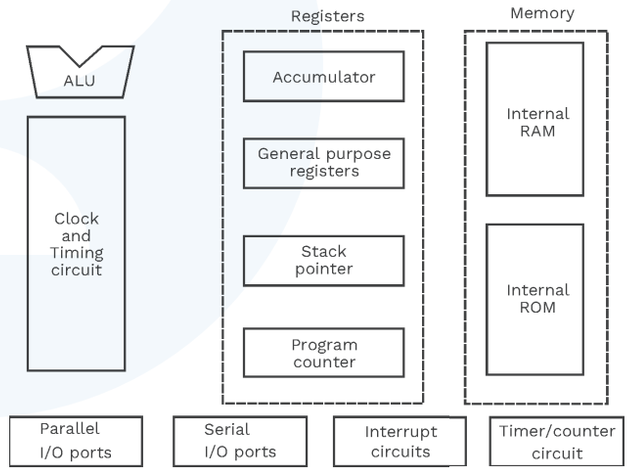


Fig. 7.2 Block Diagram of a Microcontroller

S. No.	Microprocessor	Microcontroller
1.	A microprocessor contains ALU, general purpose register, stack pointer, program counter, clock timing circuit, and interrupt circuit.	A microcontroller consists of a circuitry of microprocessor and has built-in ROM, I/O devices, timers, and counters.
2.	It has many instructions to move the data between memory and CPU.	It has one or two instructions to move the data between memory and CPU.

S. No.	Microprocessor	Microcontroller
3.	It has one or two bit handling instructions	It has many bit handling instructions.
4.	The access time for memory and I/O devices is more.	There is less access time for built-in memory and I/O devices.
5.	A microprocessor-based system requires more hardware.	A microcontroller-based system requires less hardware, reducing the PCB size and increasing the reliability.
6.	The design of a microprocessor-based system is more flexible.	The design of a microcontroller-based system is less flexible.
7.	It has single memory map for data and code.	It has separate memory map for data and code.
8.	Less number of pins are multi-functioned.	More number of pins are multi-functioned.

Table 7.1 Comparison between Microprocessor and Microcontroller

8051 Microcontroller

The 8051 microcontroller includes a family of microcontrollers that have numbers ranging from 8031 to 8751. It is available in N-channel Metal Oxide Silicon (NMOS), and Complementary and Metal Oxide Silicon (CMOS) package types.

Features of 8051 Microcontroller

The major features of 8051 microcontroller are listed below.

- 8 bit CPU
- Single 15 *V* operating supply.
- On-chip oscillator and clock circuitry.
- 128-bytes data memory on-chip
- Four register bank
- 32 bit bidirectional I/O lines
- 64-k address space for external data memory.
- 64-k address space for external program memory.
- Two 16 bit timer/counters (three on 8032/8052)
- Total five interrupts, in which three are internal interrupts and two are external interrupts
- Full duplex serial port
- Clock frequency 12 MHz

Pin Diagram of 8051 Microcontroller

Fig. 7.3 shows the pin diagram of an 8051 microcontroller and Table 7.2 shows the pin description. It is based on *N* channel HMOS technology operating at +5V. This DPI package contains a window that allows the EPROM to be erased when exposed to ultraviolet rays. During normal operation, ambient light may adversely affect the functionality of the chip. Hence, a label is pasted on the window to avoid the data to be erased under ambient light.

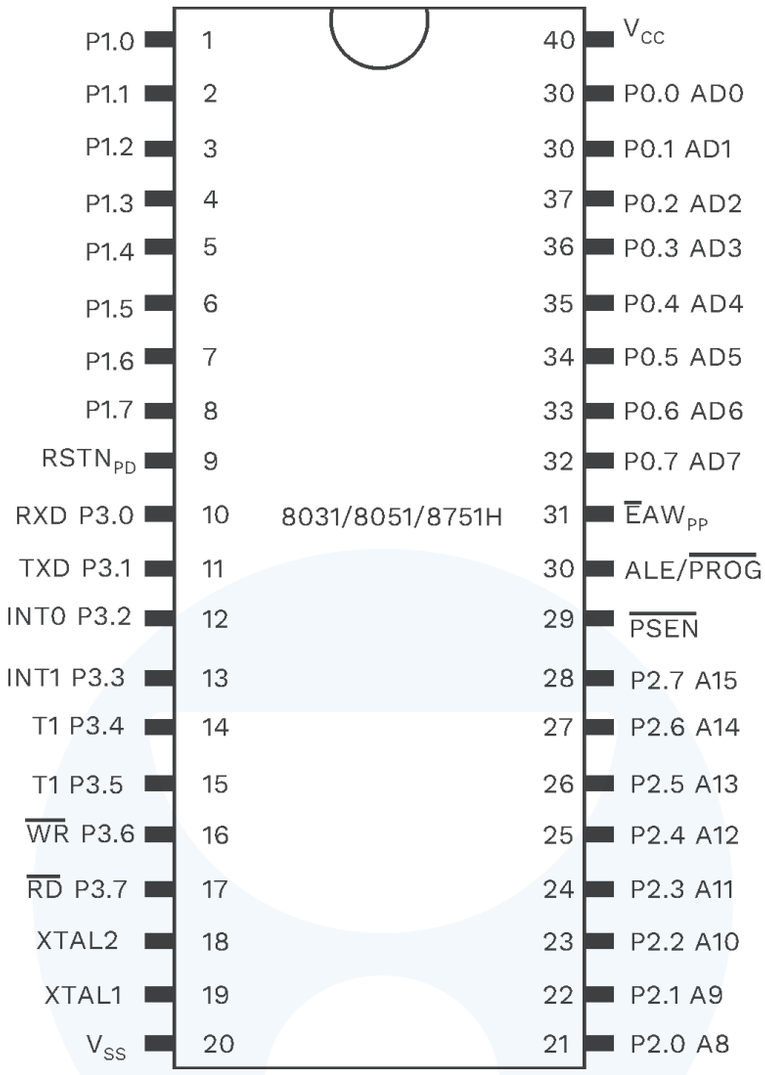


Fig. 7.3 Pin Diagram of 8051 Microcontroller

Pin No.	Pin Name	Description
32-39	Port-0	The port-0 is an 8-bit bidirectional I/O port. It works as multiplied lower order address and data bus to external memory.
1-8	Port-1	The port-1 is an 8-bit bidirectional I/O port. It receives the low order address byte during EPROM program verification.
21-28	Port-2	The port-2 is an 8-bit bidirectional I/O Port. It sends the high order address byte during access to external memory. It also receives the high order address bits during PROM programming and program verification.
20	V_{ss}	Circuit ground potential
40	V_{cc}	Supply voltage + 5V

Pin No.	Pin Name	Description
9	RST	A high on this pin resets the device when it oscillates.
30	ALE/*PROG*	The address latch enables the output for latching the low byte of the address during access to external memory. This pin is also the program pulse input (PROG) during EPROM programming.
29	*PSEN*	The program store enabled output is the read strobe to external program memory. The *PSEN* activated twice during each machine cycle is fetched from the external program memory. The PSEN Is not activated during fetches from the internal program memory.
31	*EA*/V_{PP}	External Access Enable (*EA*) is held high, the 8051H executes out of internal program memory. This pin also receives the 21V programming supply voltage (V_{PP}) during EPROM programming.
19	XTAL-1	This pin inputs to the inverting amplifier that forms the oscillator, XTAL-1 should be grounded when an internal oscillator is used.
18	XTAL-2	This pin is used as the output of an inverting amplifier that forms the oscillator to the internal clock generator.
10-17	Part-3	The port-3 is an 8-bit bidirectional I/O port with internal pull-ups. Port-3 can sink/source 4 LS TTL inputs. It also serves the functions of various special features of the 8051 family, which is given below.

Port Pin	Alternate function
P3.0	RXD (Serial input port)
P3.1	TXD (Serial output port)
P3.2	($\overline{INT_0}$) (External input)
P3.3	($\overline{INT_1}$) (External interrupt)
P3.4	T_0 (Timer/counter external input)
P3.5	T_1 (Timer/counter external input)
P3.6	($\overline{WR}^2$) (External data memory write strobe)
P3.7	$\overline{RD}$ (External data memory read strobe)

Table 7.2 Pin Description

Architecture of 8051 Microcontroller

Fig 7.4 shows the architecture of an 8051 microcontroller. It includes the following functional blocks.

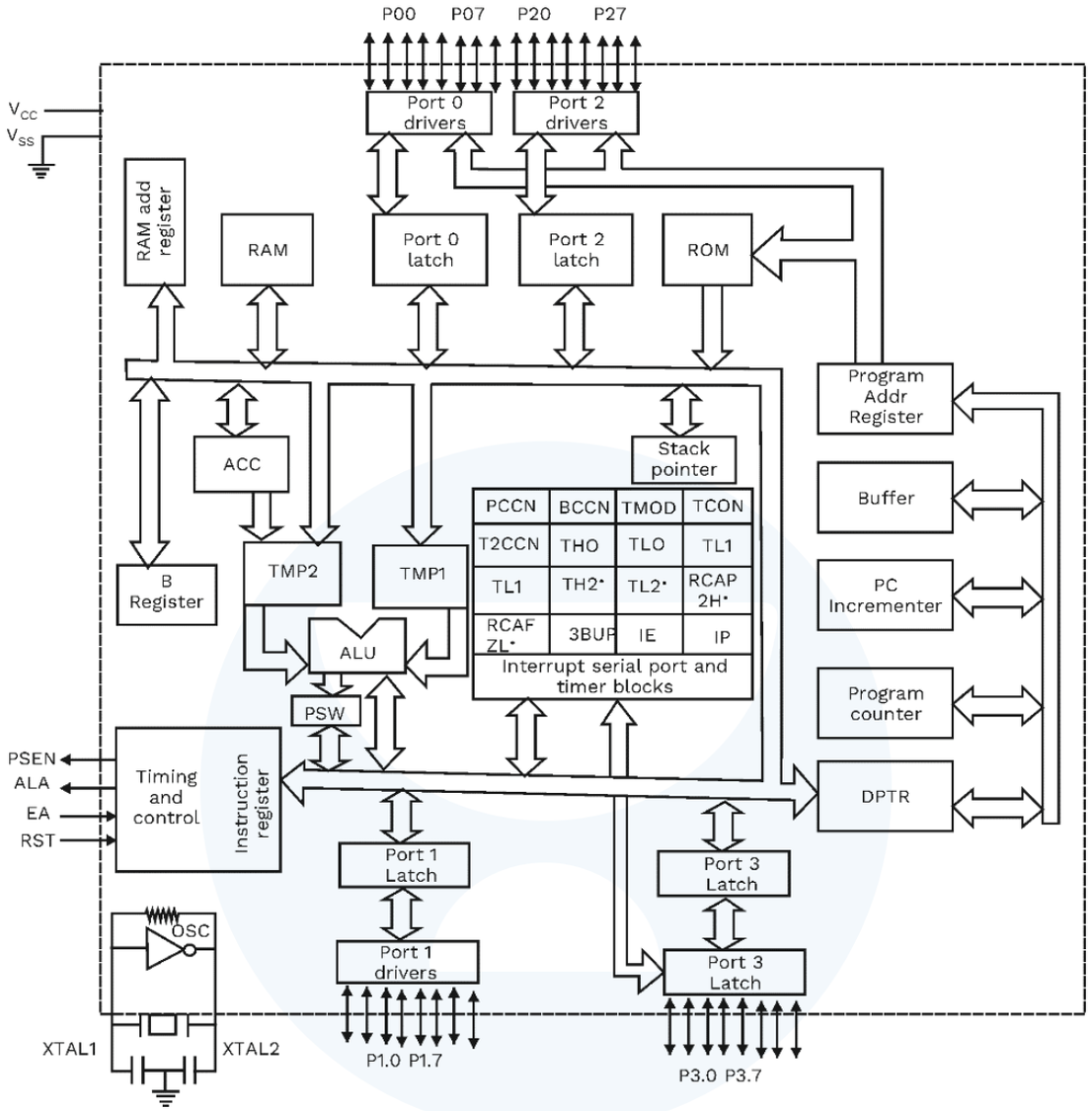


Fig. 7.4 Architecture of 8051 Microcontroller

1. The 8031/8051 has Harvard architecture, in which the same address in different memory devices or bank is used for program (or code) and data. Therefore, the architecture has two dedicated 16-bit address pointers, namely, Program Counter (PC) and Data Pointer (DPTR).
2. The PC is used as an address pointer to access program instructions. It is automatically increased after every byte of instruction fetch.
3. The DPTR is used as address pointers to read or write data in data memory. It is programmable using instructions.
4. 8031/8051 supports two memory banks of 64 kb each, one for program and the other for data.
5. The 8031/8051 has a separate 256-byte internal RAM accessed by using 8-bit address. In this 256 bytes address space, the first 128 addresses are allotted to internal RAM and the next 128 bytes are allotted to SFR. The

internal RAM/SFR can be accessed by using MOV instructions and external data memory (RAM) can be accessed by using MOVX instruction.

6. The 8051 has four 8-bit ports, namely, port 0, port 1, port 2, and port 3. Each port has a latch and driver (or buffer).
7. The port I has 110 ports and does not have any alternate function.
8. The SFRs are mapped as internal data memory. The data memory address space 80_H to FF_H are reserved for SFRs. Each register of SFR has one byte address. Some of the registers are both byte and bit addressable (the registers whose address ends with 0_H or 8_H are bit addressable).
9. The 8051 has 8-bit ALU, which performs arithmetic and logical operations on binary data. The A and B registers are used to hold the input data and the result of ALU operation. The controller will fetch the instructions one-by-one, starting from the address stored in the PC to the address stored in IR, which decodes the instructions and gives information to the timing and control unit. Using the information supplied by the IR unit, the control signals necessary for internal and external operations are generated by the timing and control unit, The 8031/8051 has an internal oscillator. So, it is sufficient if an external quartz crystal is connected for clock generation.
10. The 8031/8051 has two 16-bit programmable timer/counter 1, namely timer 1 and timer 0. In the counter mode of operation, they can count the number of high to low transitions of the signal applied to the timer pins. In the timer mode of operation, they can be independently programmed to work in any one of the four operating modes. The timer operating modes are called mode 0, mode 1, mode 2 and mode 3.
11. The 8051 family of microcontrollers has a full duplex serial port, which can be programmed to work in any one of four operating modes, namely mode 0, mode 1, mode 2, and mode 3. In mode 0, the serial port can either receive or transmit at a fixed baud rate. In mode 2, it can simultaneously transmit and receive at any one of the two selectable baud rate. In mode 1 and mode 2, it can work as a full duplex serial port with variable baud rate, which is programmed using timer 1.

Special Function Registers		Byte Address In Hexa
Symbol	**Nature**	
A or ACC	A-register or Accumulator	ED
B	13-register	F0
DPH	Data pointer higher order register	83
DPL	Data pointer lower order register	82
IE	Interrupt enable register	A8
IP	interrupt priority register	138
PO	Port 0	80
P 1	Port 1	90
P2	Port 2	A0

Special Function Registers		Byte Address In Hexa
Symbol	**Nature**	
P3	Port 3	B0
PCON	Power control register	87
PSW	Program Status Word	D0
SCON	Serial port control register	98
SBUF	Serial port data buffer	99
SP	Stack Pointer	81
TMOD	Timer/Counter mode control register	89
TCON	Timer/Counter control register	88
TLO	Timer-0 low order register	8*A*
THO	Timer-0 high order register	8*C*
TL1	Timer-1 low order register	813
TH1	Timer-1 high order register	8*D*

Table 7.3 Special Function Register

Special Function Register (SFR)

The 8031 or 8051 microcontroller is provided with 21 special function registers. These are used for selecting various programmable features of the microcontroller. The special functions of most of the SFR are distinguishable.

A and B Registers

The *A* and *B* registers are called CPU registers. They are used to hold the data for most of the CPU (ALU) operations. The size of *A* and *B* registers are 8 bit and they are mapped as on chip data memory with byte address EO_H and FO_H, respectively. These registers are also bit addressable.

Data Pointer (DPTR)

The data pointer is a 16-bit register used to hold the 16-bit address of data memory. The 16-bit data pointer can also be used as two 8-bit data pointers, namely DPH and DPL. The 8-bit data pointers are used for accessing internal RAM and SFR. The 16-bit data pointer is used for accessing external data memory.

The 8-bit data pointer DPH and DPL are mapped as internal memory with byte address 83_1 and 82_H, respectively. The contents of the data pointer are programmable using instructions.

Program Status Word (PSW)

The program status word stores the status of the result of ALU operation and some of the status of the processor by means of 1-bit status called flags. The PSW is also known as flag register.

The PSW consists of four math flags and two register bank select bits.

These flags are altered after arithmetic and logic operations, depending on the result.

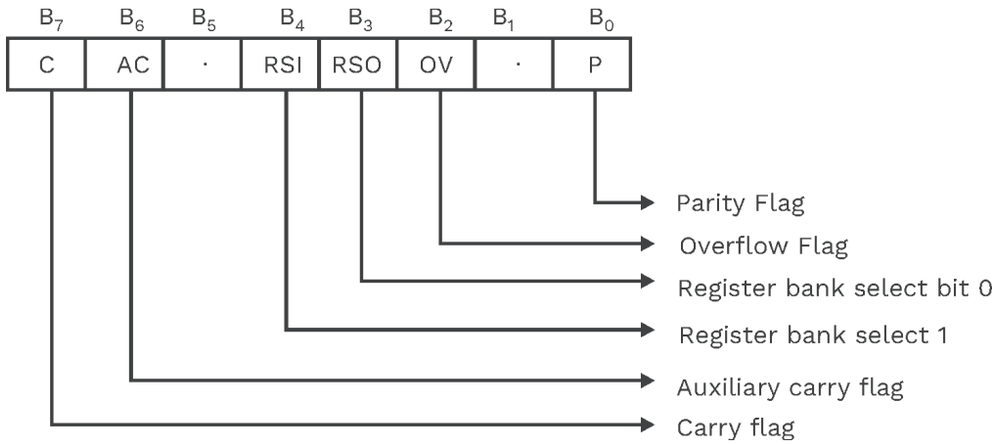


Fig. 7.5 Format of PSW of 8051 Microcontroller

The register bank select bits RS1 and RS0 are used to select any one of the four register banks of the internal RAM. At any time, the microcontroller can work with (or access) only one register bank selected by these bits.

Bank select bits		Selected register bank	Range of hexa address of the selected bank
RS1	RS0		
0	0	Bank 0	00_H–07″
0	1	Bank 1	08_H–$0AF_{H'}$
1	0	Bank 2	10_H–$17H$
1	1	Bank 3	08_H–$1F$

Table 7.4 Selection of Register Bank

Stack Pointer (SP)

The stack pointer always holds the 8-bit address on the top of the stack. The 8031/8051 microcontroller supports LIFO (Last In First Out) stack, and stack may reside anywhere in on-chip RAM (i.e., the programmer can reserve any portion of on-chip RAM as stack). After a reset, the stack pointer is initialised to 07_H. The stack can be accessed using PUSH and POP instructions. During PUSH operation, the stack pointer is automatically incremented by one and during POP operation the stack pointer is automatically decremented by one.

Power Control Register (PCON)

The PCON register is used for power control and baud rate selection. It also consists of general purpose user flags.

The controller can be driven to idle mode by setting the IDL bit of the PCON register. When idle

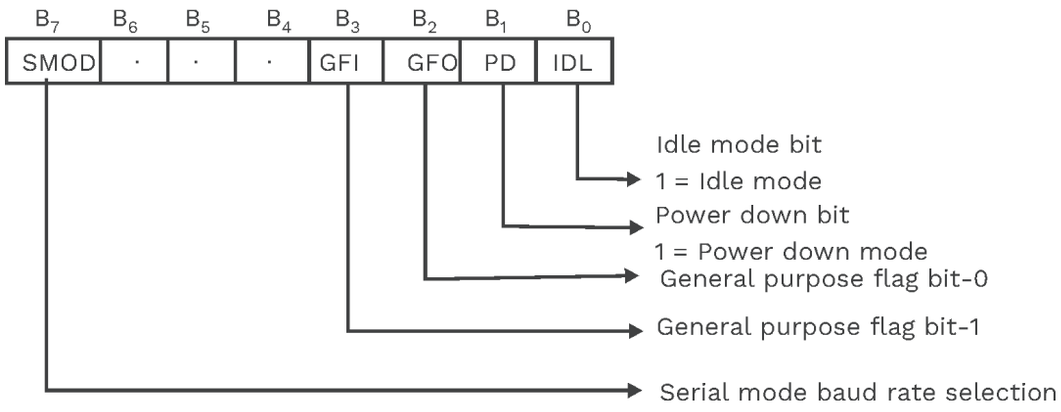


Fig. 7.6 Format of PCON Register of 8051 Family of Microcontrollers

mode is activated, the clock signal is stopped to CPU (ALU), but the clock signal is supplied to interrupt, timer, and serial port blocks.
The SMOD bit is used to decide the baud rate at serial port operating modes 1, 2 or 3. In mode 2, if SMOD = 0, then the baud rate is 1/64 of oscillator frequency; and if SMOD = 1, then the baud rate is 1132 of oscillator frequency.

Serial Data Buffer (SBUF) Register

The SBUF register is used to hold parallel data during transmission and reception. During serial reception, the serial data are received via R x D pin and converted to parallel data and stored in the receive buffer. During serial transmission, the parallel data are stored in a transmit buffer, and then, converted to serial data to transmit via T x D pin.

Serial Port Control Register (SCON)

The format of the SCON register is shown in Fig. 7.7. The bits SMO and SM 1 are used to select any one of the four operating modes for serial transmission and reception. The four modes of serial port are mode 0, 1, 2 and 3.

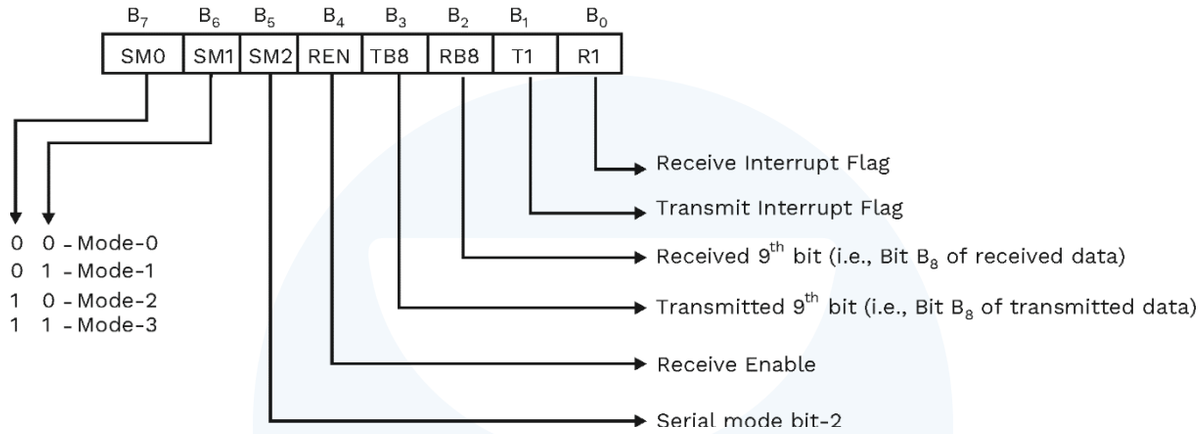


Fig. 7.7 Format of SCON Register of 8051 Family of Microcontrollers

Mode 0

In this mode, the serial port functions as a half duplex serial port with fixed baud rate. In this mode, the serial port functions as a half duplex serial port with fixed baud rate and the 8-bit serial pin during reception and transmission. The baud rate is fixed at 1/12 of the oscillator frequency.

Mode 1

In this mode, the serial port functions as a full duplex serial port with variable baud rate. In this mode, one datum consists of 10 bits that include one start bit and eight data variable baud rate. In this mode, one stop bit is stored as RB8 in the SCON register.

Mode 2

In this mode, the serial port functions as full duplex serial port with a baud rate of either 1/32 or 1/64 of the oscillator frequency. In this mode, one datum consists of 11 bits that include one start bit, eight data bit, a programmable ninth data bit, and one stop bit. During transmission, the TB8 of the SCON register is added as the ninth data bit; and during reception, the ninth data bit is stored as RB8 in the SCON register.

Mode 3

Mode 3 is the same as mode 2, except the baud rate. In Mode 3, the baud rate is variable. The baud rate depends on the value of SMOD bit in *PCON* register and timer 1 overflow rate.

Timer Mode Control (TMOD) Register

The TMOD register is used to select the operating mode and the timer or counter operation of the timers. The two timers can be independently programmed to operate in various modes. The TMOD register has two separate 2 bit field *M*0 and *M*1 to program the operating mode of timers. The operating modes of timers are mode 0, 1, 2, and 3. In all these operating modes, the oscillator clock is divided by 12 and applied as input clock to timer.

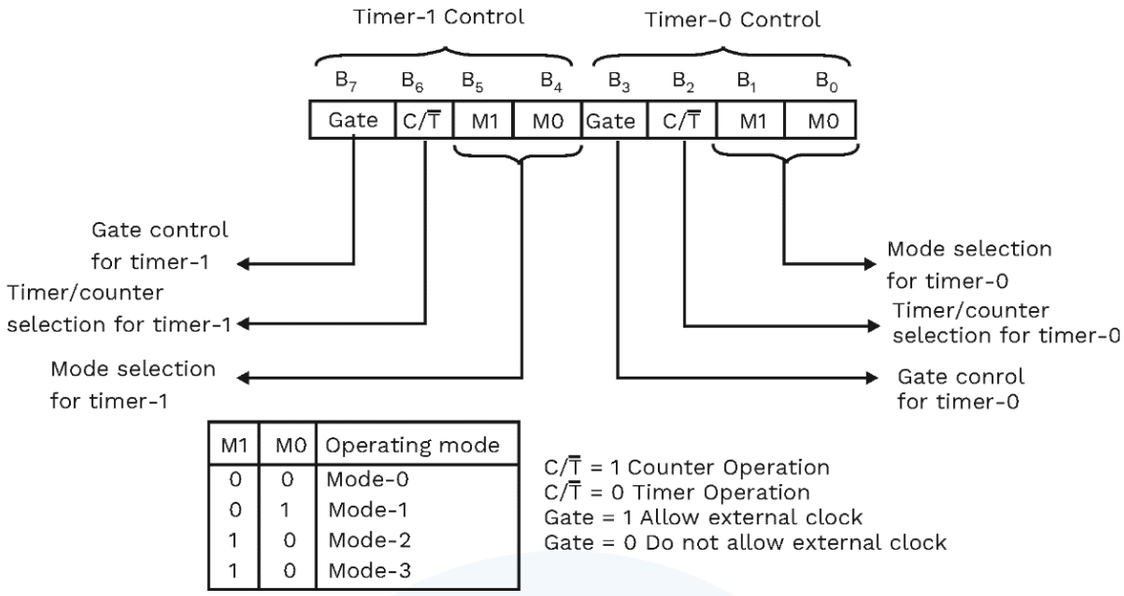


Fig. 7.8 Format of TMOD Register of 8051 Family of Microcontroller

Mode 0

In mode 0, the timer register is configured as a 13 bit register. For timer 1, the 8 bits of TH1 and lower 5 bits of TL1 are used to form a 13 bit register. For timer 0, the 8 bit THO and lower 5 bit TLO are used to form a 13 bit register.

Mode 1

Mode 1 is same as Mode 0 except the size of the timer register. In mode 1, the TH and TL registers are cascaded to form a 16 bit timer register.

Mode 2

In mode 2, the timers function as 8 bit timer with automatic reload feature.

Mode 3

In mode 3, timer 0 is configured as two separate 8 bit timers and timer 1 is stopped. In mode 3, the TLO will function as an 8 bit timer controlled by standard timer 0 control bits. The THO will function as an 8 bit timer controlled by timer 1 control bits. While timer 0 is programmed in mode 3, timer I can be programmed in mode 0, 1, or 2, and can be used for applications that do not require an interrupt.

Timer Control (TCON) Register

The TCON register consists of timer overflow flags, timer run control bits, external interrupt flags, and external interrupt type control bits. The format of TCON register is shown in Fig. 7.9.

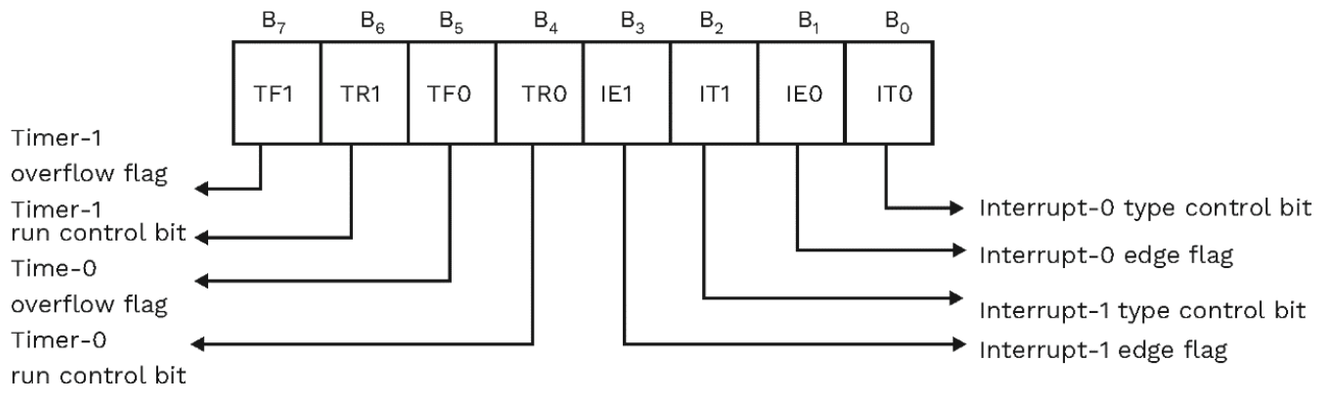


Fig. 7.9 Format of TCON Register of 8051 Family of Microcontroller

Interrupt Enable (IE) Register

The IE register is used to enable/disable the interrupts of 8051. The interrupts are recognised (or accepted) by the controller only if they are enabled. The IE register can be programmed to enable/disable all five interrupts of 8051 totally or individually. The format of IE register is shown in Fig. 7.10. The EA bit of IE register can be programmed as zero to disable all five interrupts of 8051. When EA bit is programmed as one, the interrupts are enabled provided their individual enable bits are programmed as one (the EA bit is also called global enable).

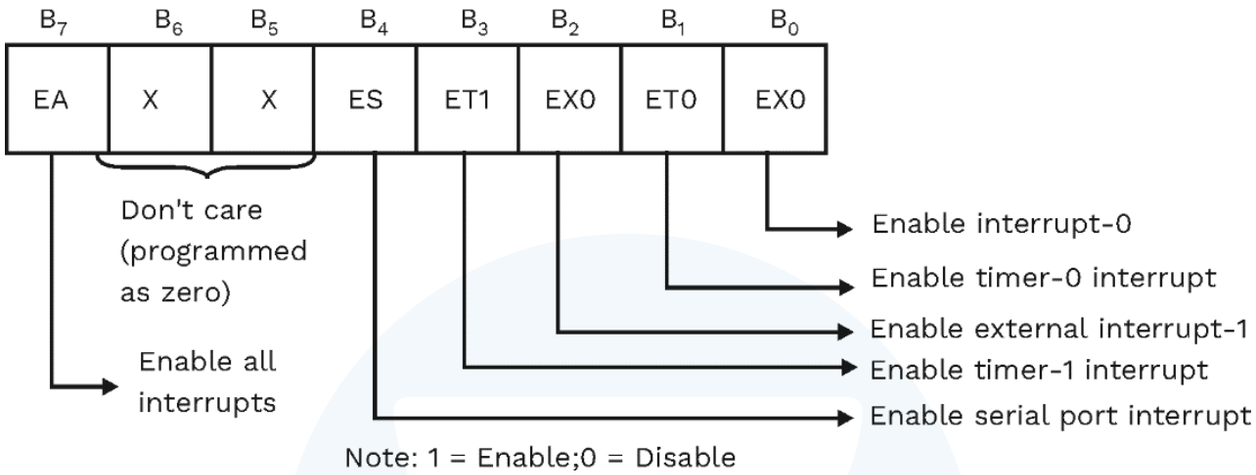


Fig. 7.10 Format of IE Register of 8051 Family of Microcontrollers

Interrupt Priority (IP) Register

The 8051 has five interrupts and the normal priority of these interrupts from highest to lowest are external interrupt 0, timer 0 interrupt, external interrupt 1, timer 1 interrupt, and serial port interrupt.

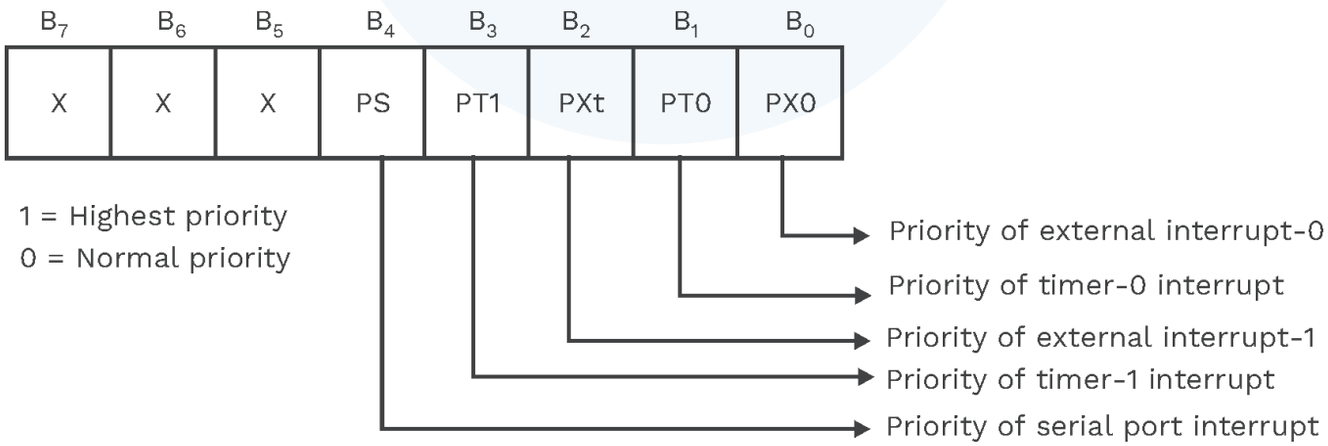


Fig. 7.11 Format of IP Register of 8051 Family of Microcontrollers

Addressing Modes

Intel 8051 has following addressing modes.

- Register addressing mode
- Direct addressing mode
- Register indirect addressing mode
- Immediate addressing mode
- Base register plus index register indirect addressing mode

Classification of Instructions

Instructions are classified as follows depending on the task they perform.

- Data transfer instructions
- Arithmetic instructions
- Logical instructions
- Branch instructions
- Stack, I/O, machine control instructions

- Data transfer instructions: in these instructions, data are transferred from the source to destination. Source can be data, contents of register, or contents of memory location. Destination can be register or memory location. They do not affect any flag.
- Arithmetic instructions: these instructions perform arithmetic operations, such as addition, subtraction, increment, and decrement. They affect the flag register.
- Logical instructions: these instructions perform logical operations, such as AND, OR, XOR, compare, rotate, and complement.
- Control transfer instructions: it includes Jump, Call, and Return instructions.

Table 8051 Instruction Set Summary

Interrupt Response Time: Refer to Hardware Description Chapter

Instructions that affect flag settings (1)							
Instructions	**Flag**			**Instructions**	**Flag**		
	C	OV	AC		C	OV	AC
ADD	C	X	X	CLRC	C		
ADDC	X	X	X	CPLC	O		
SUBB	X	X	X	ANLC, bit	X		
MUL	X	X		ANLC/bit	X		
DIV	O	X		ORLC, bit	X		
DA	O			ORLC, bit	X		
RRC	X			MOVC, bit	X		
RLC	X			CJNE	X		
SETB C	1				X		

Table 7.1

Note that operations on SFR byte address 208 or bit addresses 209-215 (i.e., the PSW or bits in the PSW) will also affect flag settings.

RN	Register R7-R0 of the currently selected Register Bank
direct	An 8 bit internal data location's address; this could be an Internal Data RAM location (0-127) or a SFR [i.e., I/0 port, control register, status register, etc. (128-255)
@Ri	An 8 bit internal data RAM location (0-255) addressed indirectly through register Rl or R0
*data	An 8 bit constant included in instruction
#data16	A 16 bit constant included in instruction

addr 16	A 16 bit destination address; used by LCALL and UMP; a branch can be anywhere within the 64K byte program memory address space
add 11	An 11 bit destination address; used by ACALL and AJMP. The branch will be within the same 2K byte page of program memory as the first byte of the following instruction
rel	Signed (two's complement) 8 bit offset byte; used by SJMP and all condition
bit	Direct addressed bit in internal data RAM or special function register

Table 7.2 8051 Instruction Set Summary

Mnemonic		Description	Byte	Oscillator Period
Arithmetic Operations				
ADD	A, Rn	Add Register to accumulator	1	12
ADD	A, direct	Add direct byte to accumulator	2	12
ADD	A, @Ri	Add indirect RAM to accumulator	1	12
ADD	A, #data	Add immediate data to accumulator	2	12
ADDC	A, Rn	Add register to accumulator with carry	1	12
ADDC	A, direct	Add direct byte to accumulator with carry	2	12
ADDC	A, @Ri	Add indirect RAM to accumulator with carry	1	12
ADDC	A, #data	Add immediate data to accumulator with carry	2	12
SUBB	A, Rn	Subtract register from accumulator with borrow	1	12
SUBB	A, direct	Subtract direct byte from accumulator with borrow	2	12
SUBB	A, @Ri	Subtract indirect RAM from accumulator with borrow	1	12
SUBB	A, *data	Subtract immediate data from Acc with borrow	2	12
INC	A	Increment Accumulator	1	12
INC	Rn	Increment Register	1	12
INC	direct	Increment direct byte	2	12
INC	@Ri	Increment direct RAM	1	12
DEC	A	Decrement Accumulator	1	12
DEC	Rn	Decrement Register	1	12
DEC	direct	Decrement direct byte	2	12
DEC	@Ri	Decrement indirect RAM	1	12

Mnemonic		Description	Byte	Oscillator Period
Arithmetic Operations (Continued)				
INC	DPTR	Increment Data Pointer	1	24
MUL	AB	Multiply A and B	1	48
DIV	AB	Divide A by B	1	48
DA	A	Decimal Adjust Accumulator	1	12
Logical Operations				
ANL	A, Rn	AND Register to Accumulator	1	12
ANL	A, direct	AND direct byte to Accumulator	2	12
ANL	A, @Ri	AND indirect RAM to Accumulator	1	12
ANL	A, #data	AND immediate data to Accumulator	2	12
ANL	direct, A	AND Accumulator to direct byte	2	12
ANL	direct, *data	AND immediate data to direct byte	3	24
ORL	A, Rn	OR Register to Accumulator	1	12
ORL	A, direct	OR direct byte to Accumulator	2	12
ORL	A, @Ri	OR indirect RAM to Accumulator	1	12
ORL	A, #data	OR immediate data to Accumulator	2	12
ORL	direct, A	OR Accumulator to direct byte	2	12
ORL	direct, #data	OR immediate data to direct byte	3	24
XRL	A, Rn	Exclusive-OR register to accumulator	1	12
XRL	A, direct	Exclusive-OR direct byte to accumulator	2	12
XRL	A, @Ri	Exclusive-OR indirect RAM to accumulator	1	12
XRL	A, #data	Exclusive-OR immediate data to accumulator	2	12
XRL	direct, A	Exclusive-OR Accumulator to direct byte	2	12
XRL	direct, *data	Exclusive-OR immediate data to direct byte	3	24
CLR	A	Clear Accumulator	1	12
CPL	A	Complement Accumulator	1	12

Mnemonic	Description	Byte	Oscillator Period
Logical Operations (Continued)			
RL A	Rotate Accumulator Left	1	12
RLC A	Rotate Accumulator Left through the Carry	1	12
RR A	Rotate Accumulator Right	1	12
RRC A	Rotate Accumulator Right through the Carry	1	12
SWAP A	Swap nibbles within the Accumulator	1	12
Data Transfer			
MOV A, Rn	Move register to accumulator	1	12
MOV A, direct	Move direct byte to accumulator	2	12
MOV A, @Ri	Move indirect RAM to accumulator	1	12
MOV A, #data	Move immediate data to accumulator	2	12
MOV Rn, A	Move accumulator to register	1	12
MOV Rn, direct	Move direct byte to register	2	24
MOV Rn, #data	Move immediate data to register	2	12
MOV direct, A	Move accumulator to direct byte	2	12
MOV direct, Rn	Move register to direct byte	2	24
MOV direct, direct	Move direct byte to direct	3	24
MOV direct, @Ri	Move indirect RAM to direct byte	2	24
MOV direct, #data	Move immediate data to direct byte	3	24
MOV @Ri, A	Move accumulator to indirect RAM	1	12
XCH A, @Ri	Exchange indirect RAM with accumulator	1	12
XCHD A, @Ri	Exchange low-order digit indirect RAM with accumulator	1	12

Mnemonic	Description	Byte	Oscillator Period
DATA TRANSFER (Continued)			
MOV @Ri, direct	Move direct byte to indirect RAM	2	24
MOV @Ri, # data	Move immediate data to indirect RAM	2	12
MOV DPTR, #datal6	Load data pointer with a 16 bit constant	3	24
MOVC A, @A + DPTR	Move code byte relative to DPTR to accumulator	1	24
MOVC AAA + PC	Move code byte relative to PC to accumulator	1	24
MOVX A, @Ri	Move External RAM (8 bit addr) to accumulator	1	24
MOVX A, @DPTR	Move External RAM (16 bit addr) to accumulator	1	24
MOVX @Ri, A	Move accumulator to External RAM (8 bit addr)	1	24
MOVX @DPTR, A	Move accumulator to External RAM (16 bit addr)	1	24
PUSH direct	Push direct byte onto stack	2	24
POP direct	Pop direct byte from stack	2	24
XCH A, Rn	Exchange register with accumulator	1	12
XCH A, direct	Exchange direct byte with accumulator	2	12

Mnemonic		Description	Byte	Oscillator Period
Boolean Variable Manipulation				
CLR	C	Clear Carry	1	12
CLR	bit	Clear direct bit	2	12
SETB	C	Set Carry	1	12
SETB	bit	Set direct bit	2	12
CPL	C	Complement Carry	1	12
CPL	bit	Complement direct bit	2	12
ANL	C,bit	AND direct bit to CARRY	2	24
ANL	C,/bit	AND complement of direct bit to Carry	2	24
ORL	C,bit	OR direct bit to Carry	2	24
ORL	C,/bit	OR complement of direct bit to Carry	2	24
MOV	C,bit	Move direct bit to Carry	2	12
MOV	bit,C	Move Carry to direct bit	2	24
JC	rel	Jump if Carry is set	2	24
JNC	rel	Jump if Carry not set	2	24
JB	bit,rel	Jump if direct Bit is set	3	24
JNB	bit,rel	Jump if direct Bit is Not set	3	24
JBC	bit,rel	Jump if direct Bit is set & clear bit	3	24
Program Branching				
ACALL	addr11	Absolute Subroutine Call	2	24
LCALL	addr16	Long Subroutine Call	3	24
RET		Return from Subroutine	1	24
RETI		Return from interrupt	1	24
AJMP	addr11	Absolute Jump	2	24
UMP	addri 6	Long Jump	3	24
SJMP	rel	Short Jump (relative addr)	2	24

Mnemonic		Description	Byte	Oscillator Period
Program Branching (Continued)				
JMP	@A + DPTR	Jump indirect relative to the DPTR	1	24
JZ	rel	Jump if accumulator is Zero	2	24
JNZ	rel	Jump if accumulator is Not Zero	2	24
CJNE	A, direct, rel	Compare direct byte to accumulator and Jump if Not Equal	3	24
CJNE	A, * data, rel	Compare immediate to accumulator and Jump if Not Equal	3	24
CJNE	Rn, # data, rel	Compare immediate to register and Jump if Not Equal	3	24
CJNE	@Hi, # dataxel	Compare immediate to indirect and Jump if Not Equal	3	24
DJNZ	Rn, rel	Decrement register and Jump if Not Zero	2	24
DJNZ	direct, rel	Decrement direct byte and Jump if Not Zero	3	24
NOP		No Operation	1	12

Table 7.3